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A comparative experiment of telluric field frequency selection method and audio-magnetotelluric method next a water well

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Abstract

The telluric field frequency selection method (TFFSM) has been proposed for more than 40 years and has achieved good exploration results in groundwater exploration. The cause of its anomaly formation has always been one of the issues that people are concerned about. This article mainly explains the causes of TFFSM anomalies through experimental detection near a known water well, and demonstrates its relevant issues in application. Firstly, the triple-frequency observation was carried out along the survey line to quickly determine the location of groundwater. Secondly, the synchronous observations of three type of frequency selectors in the case of electrode separation and electrode sharing were adopted respectively. Thirdly, the variation of TFFSM exploration results with electrode spacing (MN) was studied. Finally, audio-magnetotelluric method (AMT) measurements were taken at two points on the survey line. Through the above experiments, we obtained the potential difference (ΔV) profile curves and their pseudo-sections of TFFSM under different conditions, as well as the power spectral density curves of electric and magnetic field of AMT, and corresponding apparent resistivity curves. According to the analysis of the experimental results, it can be concluded that the existence of groundwater will cause static effects in the detection results of TFFSM. TFFSM has relatively loosen grounding requirements for electrodes, but the size of MN has a certain influence on exploration depth. From the comparison between TFFSM and AMT, the former has stronger anti-interference ability and site adaptability, and TFFSM can be applied to urban geological exploration in the future.

Highlights

- Through the telluric field frequency selection method (TEFSM) detection near a known water well, it is verified that the cause of the anomaly is static effect.
- In TEFSM exploration, the grounding mode of the electrode has little effect on the exploration results.
- In the practical application of the TEFSM, there is a certain correlation between the exploration depth and the electrode spacing (MN). The smaller MN , the smaller the observation value and the shallower the exploration depth.
- Compared with the AMT method, the TEFSM instrument is lighter, more resistant to interference and more adaptable to the site, and can achieve satisfactory results in some areas where AMT cannot work.



Keywords Frequency selection method (FSM), Audio magnetotelluric method (AMT), Static effect, Well, Groundwater

1 Introduction

The telluric field frequency selection method (TFFSM) is commonly referred to as the frequency selection method (FSM) and has a history spanning nearly five decades [24]. In prior literature, it has also been designated the telluric electric field frequency selection method (TEFSM) [30]. Developed in China, this method emulates the observational paradigm of magnetotellurics (MT), focusing exclusively on measurements of the horizontal electric field components at the Earth's surface [7, 13]. Today, TFFSM has been widely adopted in groundwater exploration. For instance, the frequency selector developed by the Hunan Puqi Geological Exploration Equipment Research Institute (Hunan Province, China) has been deployed in over 180 countries worldwide [28]. Nevertheless, certain geophysical experts remain skeptical about this method—primarily due to a lack of theoretical research on this method in the academic community.

Although both TFFSM and MT are grounded in the theory of natural electromagnetic (EM) wave propagation, their field signal acquisition strategies differ fundamentally. TFFSM employs hardware-based frequency selection: predefined detection frequencies are set in advance to target specific subsurface depths, enabling targeted suppression of noise. In contrast, MT records time-domain signals, from which electric and magnetic field components at various frequencies are subsequently extracted via spectral analysis [34]. TFFSM typically presents raw horizontal electric field measurements directly as its output, requiring minimal post-acquisition data processing, MT, by comparison, relies on observed electric and magnetic fields to compute transfer functions—including apparent resistivity, phase, and other derived parameters. Moreover, while MT entails higher operational costs and greater logistical complexity, it offers significantly deeper penetration, enabling investigation of the deep crust [19]. The operating frequencies of TFFSM typically range from 10 to 5000 Hz, with the lowest measurable frequency for certain instruments reaching as low as 2 Hz. TFFSM is primarily employed for near-surface groundwater exploration within depths of up to 500 m [22, 29], and has yielded satisfactory geological outcomes in numerous field applications [1, 9, 14–16]. Following years of practical deployment and systematic research, Yang et al. [24, 26] identified static shift—commonly associated with the natural electromagnetic (EM) method—as the predominant cause of TFFSM anomalies, although weak radar-like reflection signals may occasionally be present [6, 28]. Concurrently, they introduced the concept of the telluric magnetic field frequency selection method (TMFSM) [30].

Audio-magnetotellurics (AMT), first proposed by Berdichevski et al. [2] as an extension of MT method, shares identical observational principles and measurement parameters with MT but operates over a higher-frequency band (> 1 Hz). Consequently, AMT is generally suited for investigations targeting depths of up to approximately 2 km [5, 18]. To date, AMT has matured both theoretically and technologically and is widely applied in groundwater, mineral, and engineering geophysical surveys [8, 21, 33]. Nevertheless, compared with TFFSM, both MT and AMT are designed for deeper targets, thus, static

shift correction remains an indispensable preprocessing step in their data interpretation [20].

In this study, three distinct frequency-selection instruments for TFFSM and a conventional magnetotelluric system were jointly deployed in a groundwater prospecting experiment at a potable water well site. Through comprehensive analysis of the acquired field data, the applicability and effectiveness of these two geophysical approaches under culturally disturbed conditions are critically evaluated. Furthermore, the results substantiate that TFFSM anomalies predominantly arise from static effects. The influence of electrode spacing on TFFSM survey outcomes is also systematically examined.

2 Experiment site

The experimental site is located at the geographical coordinates $27^{\circ} 54' 9.860''$ N and $112^{\circ} 55' 7.658''$ E—specifically, the plum garden of Hunan University of Science and Technology (HNUST), Xiangtan, China (Fig. 1). A potable water well is situated at the corner of a road intersection (Fig. 1), which was identified and drilled in 1988 by faculty and students from the Department of Geology at HNUST. At that time, geophysical exploration employed the DX-1 mechanical frequency selector, developed by the Zhengzhou Geological School (Henan Province, China). The well reaches a total depth of 180.9 m and yields more than 60 t/h of groundwater. The white dashed lines in Fig. 1 denote fractures; the yellow solid line indicates the location of the survey line, and the symbol “#” marks the position of the water well. Geological interpretation indicates that the well intersects a small secondary fault trending approximately southwest (Fig. 1); the adjacent major fault strikes northwest. For several decades, this well served as the sole source of domestic water supply for the southern campus of HNUST. In recent years, however, it has been retained solely as an emergency backup, following substantial upgrades to the municipal tap-water supply infrastructure.

Photographs of the experimental site are shown in Fig. 2, with the arrow indicating the orientation of the survey line. Drilling logs reveal that the near-surface strata consist of approximately 25 m of Quaternary (Q) clay, underlain by Cretaceous (K) siltstone, glutenite, and conglomerate (Fig. 3). The static water level in the borehole is recorded at

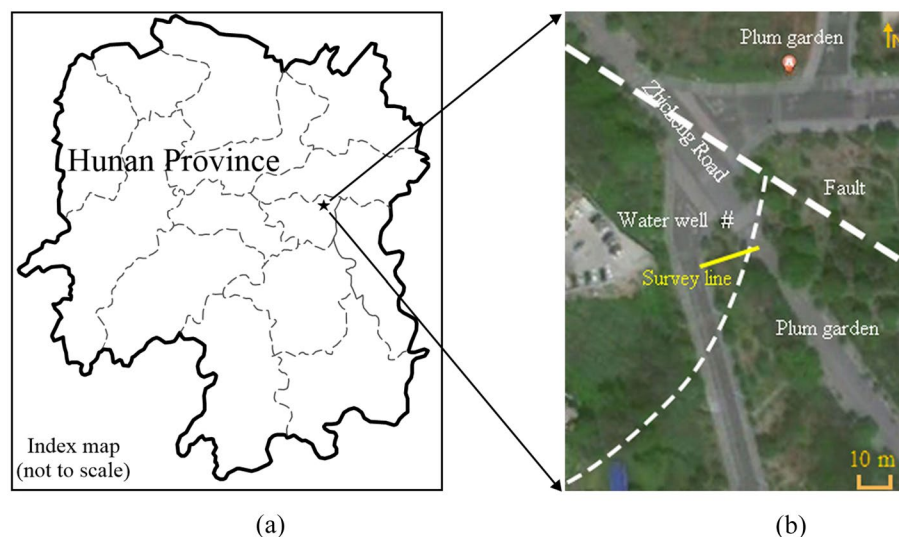


Fig. 1 Location of the experimental site



Fig. 2 Photos of the experimental location

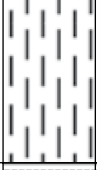
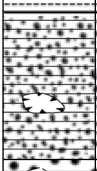
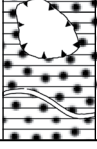
Stratigraphy	Lithology	Depth(m)	Geological characteristics
Quaternary (Q)		25.15	Brownish-red wormlike clay
Cretaceous (K)		34.20	Brown-red, thick-layered sandstone Silty-texture, blocky-structure, argillaceous cementation
		55.70	Brown and brown-red glutenite The main components of gravel are quartzite, limestone, flint, etc., with a diameter of 3~5 mm and larger ones exceeding 20 mm.
		180.90	Reddish-brown and brownish red conglomerate Siliceous and calcareous cementation The main components of gravel are limestone and quartzite, followed by sandstone, flint, sandy shale, slate, etc. The gravel is sub-round in shape, with sorting in general and a diameter of 10~80 mm.

Fig. 3 Schematic diagram of drilling column

27 m below ground surface. Two karst cavities were identified at depths of 40.3–42.3 m (height: 2.0 m) and 56.20–76.93 m (height: 20.73 m), respectively. Both cavities are filled predominantly with sand, constituting a principal aquifer for regional groundwater. Due to the presence of hardened pavement surrounding the well, the maximum feasible length of the geophysical survey line is limited to within 30 m (Fig. 2).

3 Theoretical foundations of AMT and TFFSM

3.1 AMT

The fundamental physical principle underlying both AMT and MT is identical; the primary distinction between the two methods lies in their respective observational frequency bands. AMT typically operates within the frequency range of 0.1–10⁴ Hz. As a passive electromagnetic method, AMT measures two orthogonal pairs of electromagnetic (EM) field components at the Earth's surface—namely, (E_x , H_y) and (E_y , H_x)—to infer subsurface electrical resistivity structure. Here x and y represent the north–south (N-S) and east–west (E-W) directions, respectively, while z denotes the vertical

(downward-positive) direction. Within the Cartesian coordinate system, the electromagnetic field components satisfy the following relationships:

$$\begin{bmatrix} E_x \\ E_y \end{bmatrix} = \begin{bmatrix} Z_{xx} & Z_{xy} \\ Z_{yx} & Z_{yy} \end{bmatrix} \begin{bmatrix} H_x \\ H_y \end{bmatrix} \quad (1)$$

$$[H_z] = [T_x T_y] \begin{bmatrix} H_x \\ H_y \end{bmatrix} \quad (2)$$

where E_x and E_y represent the horizontal electric field components along the x - and y -axes, and H_x and H_y denote the corresponding orthogonal magnetic field components, Z and T represent impedance and tipper tensors, respectively. The impedance can be converted into apparent resistivity ρ_{ij} and phase φ_{ij} :

$$\rho_{ij} = \frac{1}{\omega \mu_0} |Z_{ij}|^2 \quad (3)$$

$$\varphi_{ij} = \tan^{-1} \left(\frac{\text{Im}(Z_{ij})}{\text{Re}(Z_{ij})} \right) \quad (4)$$

where ω denotes angular frequency, μ_0 is magnetic permeability, ij represents any combination of x and y , and Im and Re represent the imaginary and real parts of impedance, respectively. The results computed from Eq. (3) are conventionally referred to as Cagniard resistivities [4].

All electromagnetic phenomena obey Maxwell's equations. Under the quasi-static approximation—where displacement current is negligible—EM wave propagation in conductive subsurface media is governed by the skin effect. The theoretical skin depth δ is defined as the depth at which the amplitude of an EM field decays to $1/e$ of its surface value, and is expressed as:

$$\delta \approx 503 \sqrt{\rho/f} \quad (5)$$

where ρ is the bulk resistivity of the medium (in $\Omega \cdot \text{m}$), f is the signal frequency (in Hz), and δ is given in meters.

It is critical to note that the skin depth δ does not equate to the effective exploration depth of EM methods. Due to factors such as source geometry, noise contamination, and lateral heterogeneity, the practical resolution limit—the effective exploration depth—is consistently shallower than δ and is commonly estimated as a fraction of δ (i.e., $k \cdot \delta$, where $k < 1$ is an empirically derived attenuation factor). Nevertheless, δ remains a first-order indicator of depth sensitivity: for a given geological setting, lower frequencies yield larger skin depths and thus greater potential penetration, whereas higher frequencies correspond to shallower investigation depths—as directly implied by Eq. (3).

3.2 TFFSM

TFFSM primarily exploits the static effects—also termed static shifts—of natural electromagnetic (EM) methods to detect shallow electrical anomalies [26]. In layered Earth media, EM fields exhibit predominantly horizontal distribution, and induced currents flow preferentially in the horizontal plane. When a near-surface electrically heterogeneous body is present, it perturbs the ambient EM field, generating localized regions of intensified or attenuated field distortion. This distortion disrupts the uniformity of

horizontal current density in the shallow subsurface, thereby inducing abrupt increases or decreases in the observed surface electric field components [10].

From a geophysical perspective, a primary electric field $\mathbf{E}$ oriented normal to the boundary of a heterogeneous body induces interfacial charge accumulation—termed bulk charge. Based on Gauss's law, the current continuity equation, and the quasi-static approximation, the bulk charge density q_v in a conductive medium is expressed as [11]:

$$q_v = \frac{\varepsilon_0}{\sigma} \mathbf{E} \cdot \nabla \sigma \quad (6)$$

where σ denotes the electrical conductivity of the medium, and ε_0 is the vacuum permittivity, $\varepsilon_0 = 8.85 \times 10^{-12}$ F/m. The additional electric field $\mathbf{E}_s$ generated by this bulk charge is given by [17]:

$$\mathbf{E}_s = -\frac{\partial \mathbf{A}_s}{\partial t} - \nabla \varphi \quad (7)$$

where $\mathbf{A}_s$ and φ represent the vector potential of the secondary magnetic field and electrostatic potential arising from boundary charges, respectively, and t is the time. When the physical dimensions of the heterogeneous body are significantly smaller than the EM skin depth, the time-derivative term in Eq. (7) becomes negligible relative to the gradient term; consequently, $\mathbf{E}_s$ is frequency-independent. Applying Coulomb's law, Eq. (7) can be reformulated as:

$$\mathbf{E}_s \approx -\nabla \varphi = -\nabla \int \frac{q_s}{4\pi\varepsilon_0 |\mathbf{r}|} ds = -\frac{1}{4\pi} \int_V \frac{\mathbf{E} \cdot \nabla \sigma}{\sigma |\mathbf{r}|^2} \mathbf{r}_0 dV \quad (8)$$

where ds is the surface charge element, $\mathbf{r}$ is the position vector from the volume element dV to the observation point, and $\mathbf{r}_0$ is its unit vector.

Equation (8) reveals that $\mathbf{E}_s$ is directly proportional to the primary electric field $\mathbf{E}$ and remains in phase with it. Thus, the influence of a near-surface heterogeneity scales the primary electric field via a multiplicative factor—termed the static shift factor. From the perspective of electric flux, the fundamental cause of static shifts is the localized distortion of electric flux due to near-surface charge accumulation (i.e., a non-uniform source term), rather than the electromotive force induced by time-varying magnetic flux. At sufficiently low frequencies and for heterogeneities whose size is small compared to the EM wavelength, the magnetic field remains unaffected by interfacial charge accumulation [23]. As only the electric field is modified, the apparent resistivity derived from frequency-domain EM measurements undergoes a frequency-independent, multiplicative shift in logarithmic space.

This theoretical framework indicates that the static offset in frequency-domain EM arises from distortion of the horizontal potential difference between measurement electrodes—induced either by near-electrode surface conductors or by shallow subsurface heterogeneities. Although static offset is conventionally treated as an unavoidable source of interference—particularly when targeting deeper structures [3]—TFFSM deliberately leverages this effect. Theoretical modeling demonstrates robust correlations between static shift characteristics and the geometric, electrical, and geological properties of shallow heterogeneities [26]. By quantitatively analyzing the spatial and amplitude variations of static offsets, TFFSM facilitates the inference of shallow target geometry,

conductivity contrast, and lateral extent, thereby enabling high-resolution detection of near-surface electrical anomalies.

4 Instruments and parameters

Three frequency selection instruments—namely, the PQWT-TC150, PQWT-TC300, and PQWT-TC1200—were employed in the TFFSM experiments. Their respective maximum detection depths are approximately 150 m, 300 m, and 1200 m. The PQWT-TC150 operates across 33 discrete frequencies within the range of 170–5000 Hz; the PQWT-TC300 covers a broader band of 12–5000 Hz using 40 discrete frequencies; and the PQWT-TC1200 supports three configurable maximum detection depths—600 m, 900 m, and 1200 m. For this study, the 900-m configuration was selected, utilizing 48 discrete frequencies spanning 8–5000 Hz. All instruments exhibit a measurement resolution of 0.001 mV and are rated for operational use within an ambient temperature range of -20 to $+50$ °C.

The experimental procedure comprised the following steps:

- (1) A triple-frequency (25 Hz, 67 Hz, and 170 Hz) potential difference survey was conducted along the designated survey line to preliminarily identify zones exhibiting relatively low apparent resistivity anomalies—indicative of potential subsurface groundwater presence. Data acquisition was performed using the PQWT-TC150, with a station interval and electrode spacing (MN) of 1 m and 10 m, respectively.
- (2) Three frequency selectors simultaneously acquire data at the same measurement point using their respective electrode pairs, and then collectively advance along the survey line (Fig. 4a, c).
- (3) Three frequency selectors share a single pair of electrodes to simultaneously acquire data at the same measurement point, and then collectively advance along the survey line (Fig. 4b, c).
- (4) Repeated measurements are conducted along the survey line using three distinct electrode spacings ($MN = 2, 6$, and 10 m). Data acquisition is performed using the PQWT-TC300 frequency selector. The shielding receiving cable supplied by the

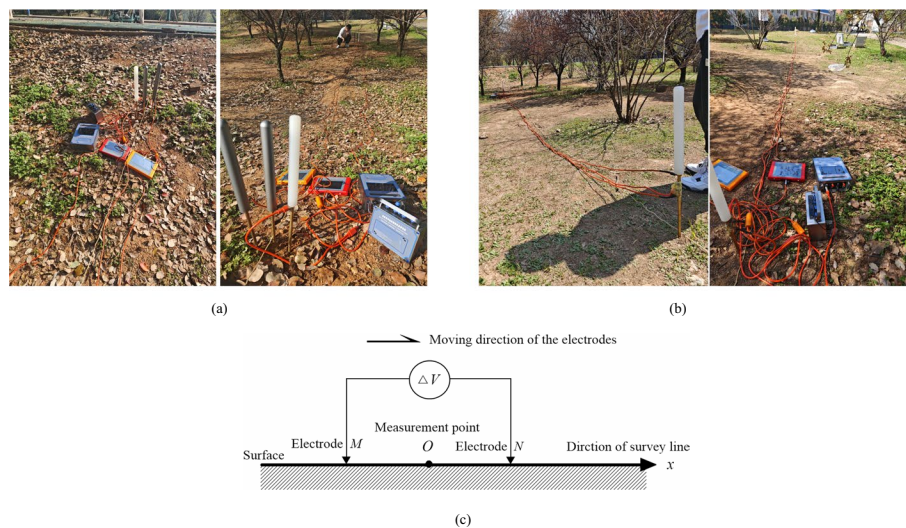


Fig. 4 Experimental photos and schematic configurations of TFFSM. **a** Using three pairs of electrodes; **b** sharing a pair of electrodes; **c** schematic illustration of the TFFSM configuration

instrument manufacturer is 10 m in length; consequently, the maximum feasible MN spacing in this experiment is limited to 10 m.

- (5) AMT measurements are carried out along the survey line (Fig. 5). Due to spatial constraints at the experimental site, the AMT survey records only the electric field components at two locations spaced 5 m apart along the line; magnetic field probes remain stationary at a fixed observation point.

In the experiments described in Steps 2–4 above, multi-frequency observations were employed, with an electrode separation (MN) of 10 m and a station interval of 1 m. In Step 5, the AMT measurements were conducted using the Surface Electromagnetic Prospecting (SEP) system—developed by the Institute of Geology and Geophysics, Chinese Academy of Sciences—with a 20-m electrode spacing for electric field component measurement. AMT data processing was performed using the SSMT-2000 software package provided by Phoenix Geophysics Co., Ltd.

5 Results and discussion

5.1 TFFSM

A triple-frequency acquisition mode (170 Hz, 67 Hz, and 25 Hz) was initially adopted to record the potential difference (ΔV) along the survey line and to identify approximate anomaly locations (Fig. 6). At each measuring station, the typical acquisition time was approximately 3 s—corresponding to rapid, preliminary reconnaissance in practical field surveys.

The three operating frequencies of the frequency selector—25 Hz, 67 Hz, and 170 Hz—are pre-set by the manufacturer. These frequencies were selected for the following reasons [13]: first, 25 Hz corresponds to one of the Schumann resonance frequencies generated by electric field resonance between the Earth's surface and the ionosphere; second, 67 Hz and 170 Hz are associated with natural magnetotelluric field fluctuations caused by distant lightning and other atmospheric activities; third, these frequencies avoid odd multiples of 50 Hz, thereby minimizing interference from power-line noise commonly present in industrial environments.

According to the exploration results obtained using the TFFSM triple-frequency mode, an anomalous zone is identified at 8–9 m along the survey line, characterized by a relatively pronounced low-potential anomaly (Fig. 6). To investigate this anomaly in greater detail, additional multi-frequency surveys were conducted. The sampling frequency range spans 8–5000 Hz. Figures 7, 8 and 9 present the simultaneously acquired data from three instruments, each employing its designated electrode configuration (Fig. 4a).



Fig. 5 Photos of AMT experiment

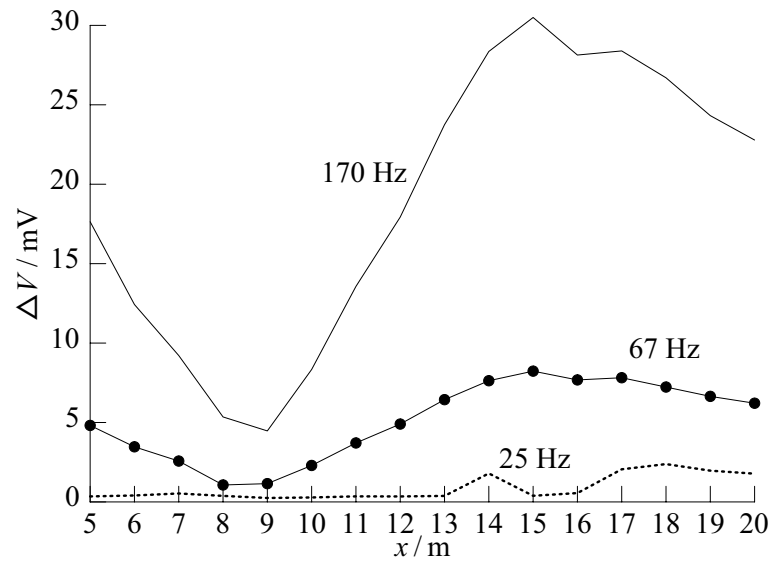


Fig. 6 TFFSM results on the survey line using three different frequencies as solid, dot-solid and dashed lines with frequency values as 170 Hz, 67 Hz and 25 Hz, respectively

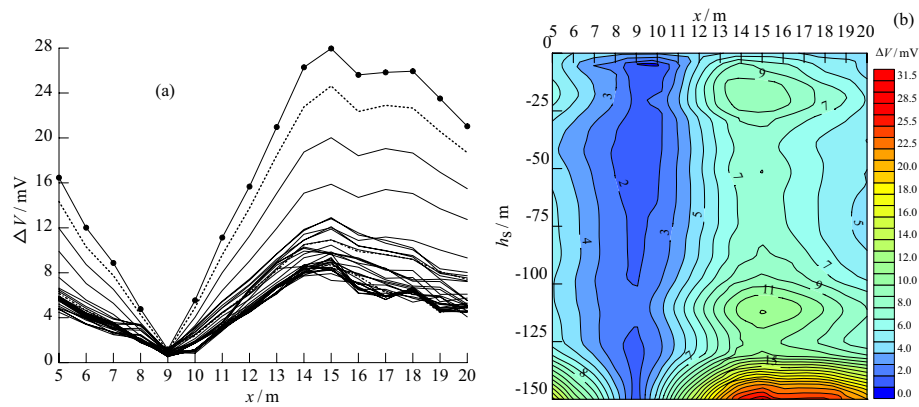


Fig. 7 The 33-frequencies detection results of TC150 using separated electrodes. **a** Profile curves of potential difference ΔV ; **b** pseudo-section of ΔV

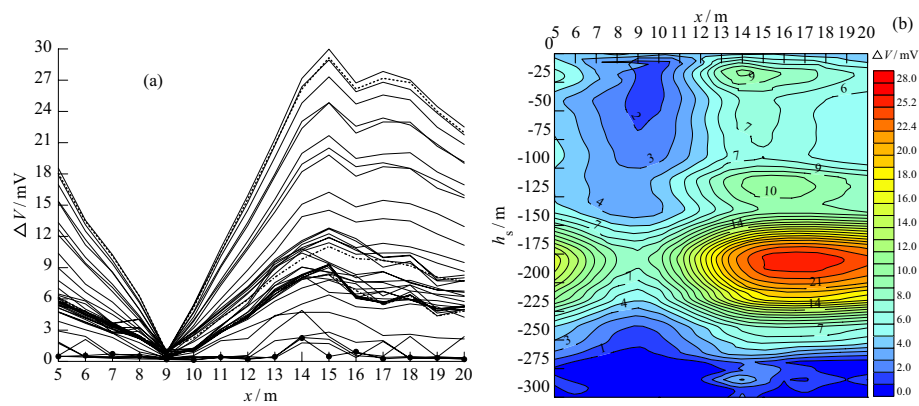


Fig. 8 The 40-frequencies detection results of TC300 using separated electrodes. **a** Profile curves of potential difference ΔV ; **b** pseudo-section of ΔV

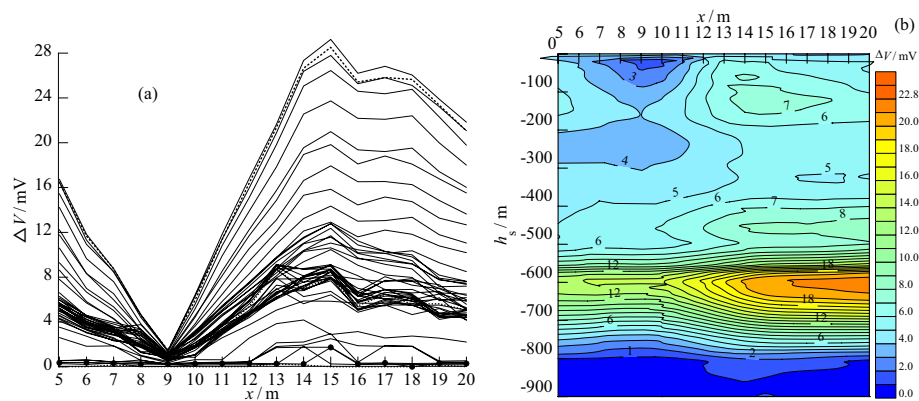


Fig. 9 The 48-frequencies detection results of TC1200 using separated electrodes. **a** Profile curves of potential difference ΔV ; **b** pseudo-section of ΔV

At the 9-m position along the survey line, a distinct low-potential-difference anomaly is consistently observed across all three profile curves (Figs. 7a, 8a and 9a), thereby corroborating the reliability of the initial triple-frequency detection results (Fig. 6). Notably, the temporal synchrony of curve fluctuations is superior for the TC150 profile (Fig. 7a) compared with those of TC300 and TC1200 (Figs. 8a and 9a). This difference arises from the distinct frequency bands employed by each instrument: asynchronous behavior is predominantly observed in the lower-frequency components. The dotted–solid lines in Figs. 7a, 8a, and 9a denote the detection results corresponding to the lowest operational frequency of each instrument. Specifically, the minimum frequency of the TC150 frequency selector exceeds 100 Hz, whereas those of TC300 and TC1200 are approximately 12 Hz and 8 Hz, respectively. As established in prior studies [12], high-frequency electromagnetic (EM) signals exhibit shallow penetration depths, whereas low-frequency signals achieve greater depth of investigation. However, deep-penetrating low-frequency signals are inherently weaker and more susceptible to cultural noise. Consequently, the observed asynchrony in the profile curves (Figs. 8a and 9a) is likely attributable to environmental interference.

According to the EM wave skin depth formula and the empirically derived exploration depth coefficient [1], the corresponding pseudo-sections of ΔV were generated (Figs. 7b, 8b and 9b). A “noodle-like” anomaly is observed between the 8th and 11th meters along the survey line in Figs. 7b, 8b, and 9b—this feature reflects the static shift effect. Concurrently, it also indicates the presence of shallow, low-resistivity, water-saturated geological bodies. Although the acquisition frequencies differ across the three datasets, the manifestation of the static effect varies in detail; notably, the effect is most pronounced in Fig. 7b. In Fig. 9b, the larger exploration depth necessitates lower frequency sampling, resulting in a comparatively weaker expression of the static effect. Taking the 3-mV contour as an example, its length along the downward extension direction in the pseudo-sections of TC150 (Fig. 7b), TC300 (Fig. 8b), and TC1200 (Fig. 9b) exceeds 150 m, is approximately 100 m, and is less than 100 m, respectively.

Figures 10, 11 and 12 present the synchronous acquisition results obtained using three kinds of frequency selectors sharing a single pair of electrodes (Fig. 4b). Under this configuration, the profile curve characteristics closely resemble those shown in Figs. 7, 8 and 9; notably, the low-frequency signal variations exhibit greater regularity (Figs. 10a, 11a and 12a). This enhanced consistency is likely attributable to reduced external EM

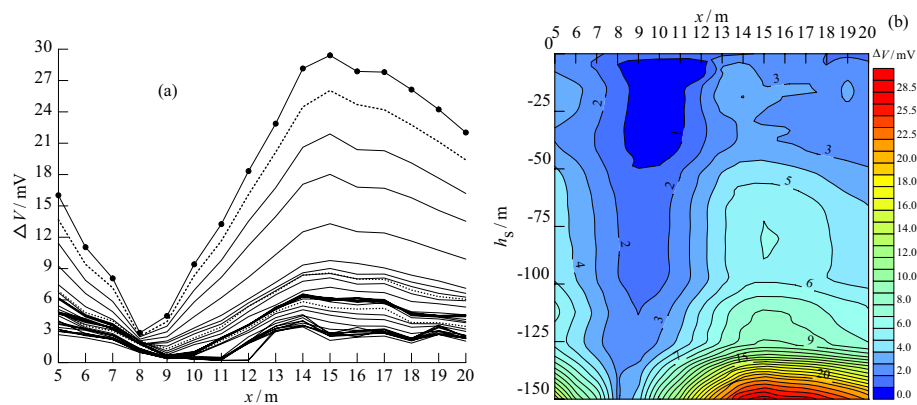


Fig. 10 The 33-frequencies detection results of TC150 using sharing electrodes. **a** Profile curves of potential difference ΔV ; **b** pseudo-section of ΔV

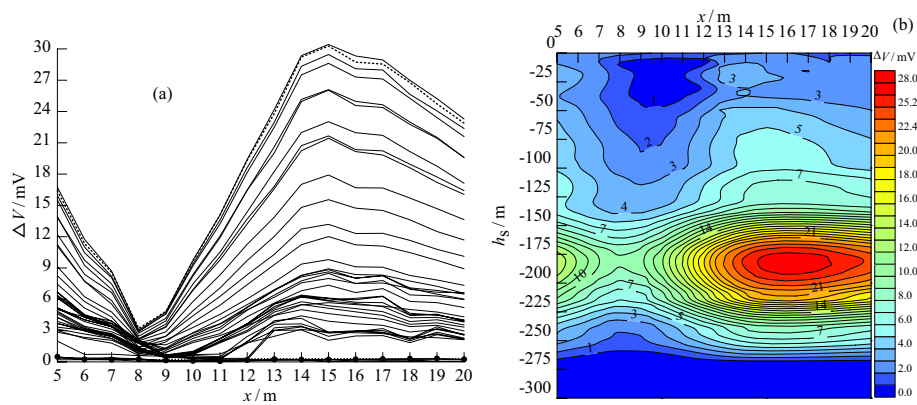


Fig. 11 The 40-frequencies detection results of TC300 using sharing electrodes. **a** Profile curves of potential difference ΔV ; **b** pseudo-section of ΔV

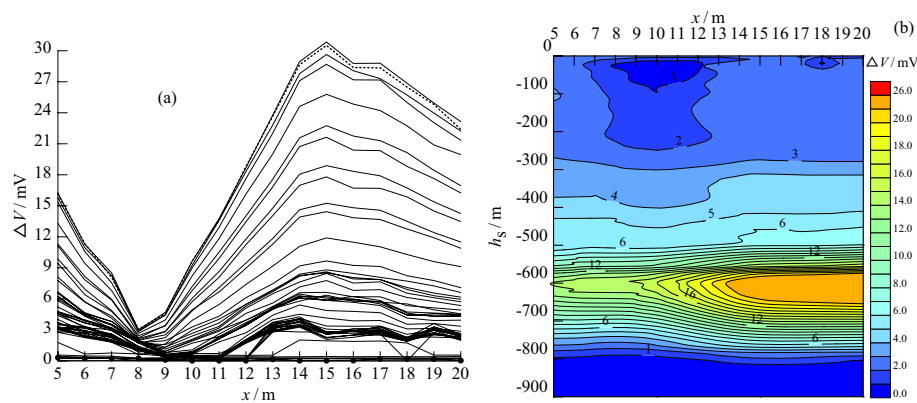


Fig. 12 The 48-frequencies detection results of TC1200 using sharing electrodes. **a** Profile curves of potential difference ΔV ; **b** pseudo-section of ΔV

interference during the acquisition window, as well as the mitigation of inter-station variability arising from electrode grounding differences. Prior studies have demonstrated that the TFFSM imposes relatively relaxed requirements on electrode grounding quality [25, 26, 31], thus, the authors attribute the observed improvement primarily to the

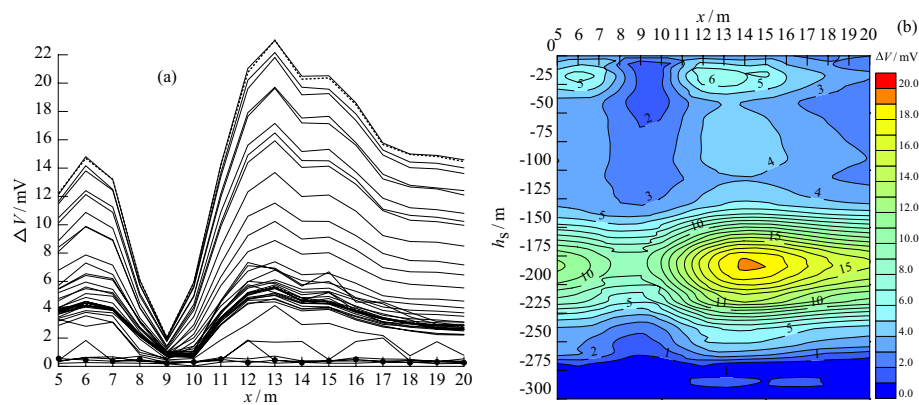


Fig. 13 The detection results of TC300 when MN is 6 m. **a** Profile curves of potential difference ΔV ; **b** pseudo-section of ΔV

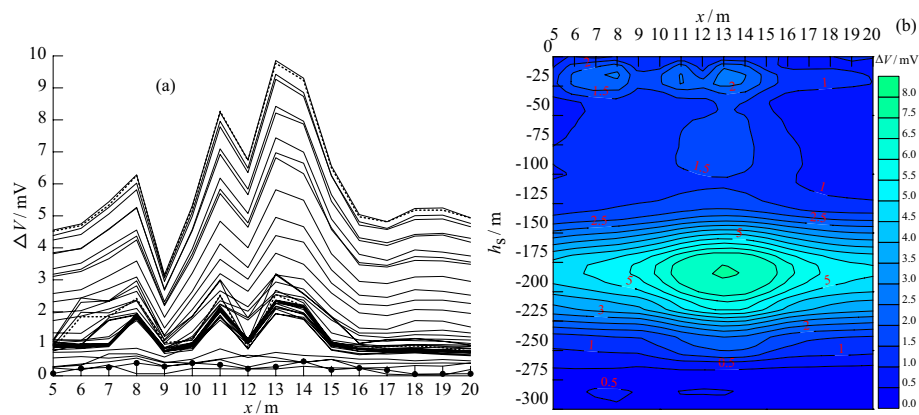


Fig. 14 The detection results of TC300 when MN is 2 m. **a** Profile curves of potential difference ΔV ; **b** pseudo-section of ΔV

former factor—i.e., lower ambient noise levels. Furthermore, aside from minor amplitude variations induced by diurnal effects, the static shift characteristics evident in Figs. 10b, 11b and 12b are consistent with those in Figs. 7b, 8b and 9b, confirming the stability of TFFSM observations.

Next, we examine the influence of electrode spacing (MN) on the observation results of the TFFSM. In all preceding experiments, MN was fixed at the commonly adopted value of 10 m; therefore, only the detection results for $MN = 6$ m and $MN = 2$ m are presented below (Figs. 13 and 14). As MN decreases, the measured potential differences (ΔV) decline markedly: the peak ΔV values in Figs. 8, 10, 13 and 14 are 29.979 mV, 30.428 mV, 23.041 mV, and 9.855 mV, respectively. Concurrently, the width of the low-potential anomaly—observed at the 9-m position along the survey line—progressively narrows (Figs. 8a, 11a, 13a and 14a), reflecting increasingly refined contour-line resolution (Fig. 14b).

In addition, the relatively low-resistivity anomalous conductive layer apparent at depth in the pseudo-sections—similar to those shown in Figs. 11b and 12b—is unreliable and arises from an inherent limitation of the method. Specifically, as exploration depth increases, the measured signal attenuates and the operating frequency decreases; consequently, in practical applications, it becomes difficult to unambiguously interpret the deepest low-potential anomaly as a true low-resistivity feature.

When $MN = 2$ m, the static shift effect induced by shallow groundwater (as observed in Fig. 14b) vanishes. Correspondingly, the observed potential difference values decrease, and the entire pseudo-section is predominantly characterized by low-values (blue) anomalies (Fig. 14b). This observation indicates that TFFSM results are sensitive to electrode spacing (MN). To detect deeper targets using TFFSM, a larger MN spacing may be required. The underlying cause likely lies in the burial depth of the water-bearing cavities—exceeding 40 m—combined with an excessively small MN spacing of only 2 m. Under these conditions, the potential difference induced by the deep cavities falls below the detectable threshold within the 2-m electrode interval. Moreover, reducing MN exacerbates instability—particularly in the low-frequency portion—of the acquired data. A careful comparison of the profile curves for $MN = 10$ m (Fig. 11a), $MN = 6$ m (Fig. 13a), and $MN = 2$ m (Fig. 14a) reveals that decreasing MN leads to an increasing number of profile curves converging toward the horizontal axis (0 mV), and these curves behavior become increasingly erratic. Further examination confirms that the frequencies associated with these anomalous profile curves are generally below ~ 20 Hz. Based on these experimental findings, it is unreasonable for frequency selector manufacturers to prescribe a fixed MN spacing (e.g., 10 m or 20 m) across all applications. Instead, users should adaptively select MN according to the expected burial depth of the target during practical surveys.

In the application of TFFSM, the natural EM field exhibits diurnal variation; however, its overall amplitude remains relatively stable between 08:00 and 18:00 daily [32]. Moreover, TFFSM enables rapid data acquisition—in practice, the observation time for a single survey line is typically short. Consequently, diurnal variations are routinely neglected in routine processing. Thus, spatial variations in the electric field components along the survey line direction directly reflect lateral and vertical changes in subsurface electrical properties. Based on over four decades of practical application experience with TFFSM, the influence of diurnal variation on exploration results is negligible and can be entirely disregarded [28].

5.2 AMT

To further investigate the influence of the surrounding environment on natural-source EM methods, AMT field experiments were conducted at the test site (Fig. 5). A vector measurement configuration was employed, simultaneously recording orthogonal electric field components (E_x and E_y) and orthogonal magnetic field components (H_x and H_y). Non-polarizing electrodes were used to measure the electric field components. Due to spatial constraints at the site, electric field measurements were restricted to two positions along the survey line— $x = 10$ m and $x = 15$ m—while the magnetic field sensors remained fixed at their initially determined locations throughout the observation period.

For direct comparison with the TFFSM results, only the power spectral density (PSD) curves of the electric field component E_x and the PSD curves of the magnetic field component H_y are presented for the AMT dataset (Fig. 15). Both the PSD (PSD_{E_x} and PSD_{H_y}) curves exhibit pronounced fluctuations and discontinuities across frequency, indicating significant contamination from ambient EM interference. The experimental site is located within a university campus, where multiple concurrent interference sources are present—including vehicular and pedestrian traffic on adjacent roads, nearby high-voltage power transmission lines, and various electrical appliances and infrastructure. In practice, such a highly disturbed EM environment renders AMT unsuitable for routine geophysical

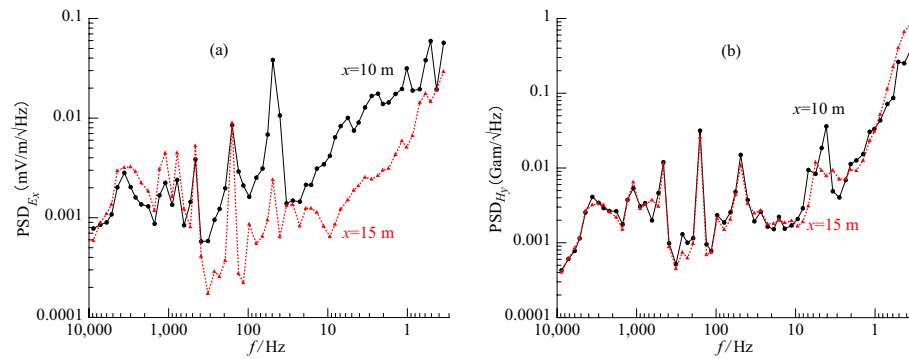


Fig. 15 Power spectral density curves of **a** electric field component E_x and **b** magnetic field component H_y

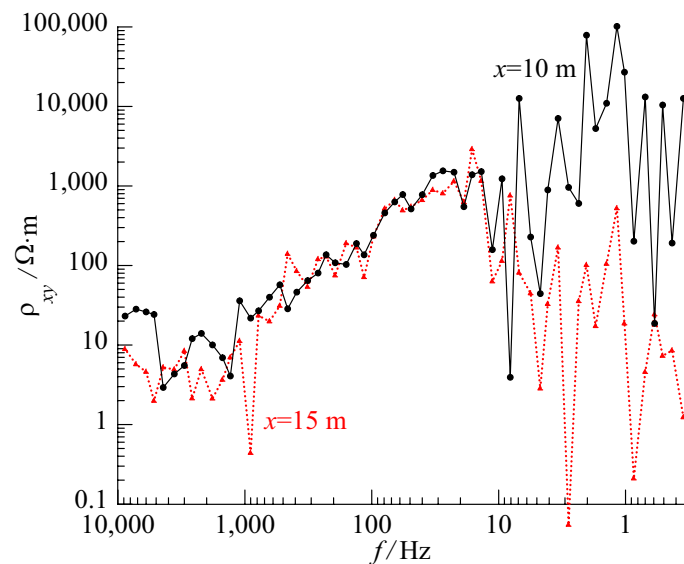


Fig. 16 The apparent resistivity results of AMT at 10 m and 15 m

exploration. In contrast, TFFSM demonstrates robust performance near high-voltage transmission lines: when the survey line is oriented parallel to the transmission lines, high-quality subsurface imaging can be achieved [27].

The variation trends of the PSD_{E_x} curves at the measurement stations located at $x = 10$ m and $x = 15$ m are generally consistent (Fig. 15a); the observed differences are likely attributable primarily to geological heterogeneity between the two sites. In contrast, the PSD_{H_y} curves for the two stations overlap almost perfectly (Fig. 15b)—this is due to the fixed position of the magnetic probe. Minor discrepancies between the two curves arise mainly from variations in ambient noise levels during data acquisition and diurnal geomagnetic fluctuations.

The AMT measurements of the electric and magnetic field components exhibit poor reliability (Fig. 15); consequently, the derived apparent resistivities fail to accurately represent the true subsurface electrical structure at the survey stations (Fig. 16). Figure 16 presents the apparent resistivity (ρ_{xy}) curves computed from E_x and H_y , where the black dot–solid line and red triangle–dashed line corresponds to $x = 10$ m and $x = 15$ m, respectively. These ρ_{xy} curves display pronounced fluctuations and are severely contaminated by strong noise—particularly at frequencies below 10 Hz (Fig. 16).

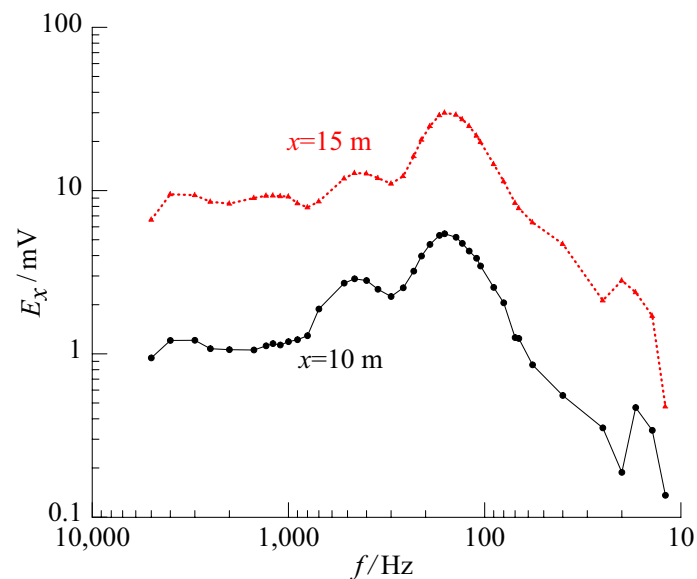


Fig. 17 The 40-frequencies detection results E_x of TC300 at 10 m and 15 m

We now revisit the detection results obtained using the PQWT-TC300 frequency selector at $x = 10$ m and $x = 15$ m, as shown in Fig. 8. Figure 17 plots the frequency-dependent variations of the TFFSM E_x component at the two survey stations. Unlike the highly erratic behavior observed in Figs. 15 and 16, the curves in Fig. 17 exhibit relatively smooth and gradual changes. This suggests that the frequency selector is significantly less susceptible to external electromagnetic interference. Its superior anti-interference performance is likely attributable to its unique signal acquisition methodology. As introduced previously, unlike AMT—which acquires data in the time domain—the TFFSM instrument determines its sampling frequencies through dedicated hardware configurations. Moreover, noise suppression is implemented directly at the hardware level, which contributes to the stability of the acquired signals and enables rapid data acquisition. Therefore, in the future development of AMT instrumentation, we propose adopting a frequency-selective data acquisition approach—namely, sequentially sampling signals at discrete frequencies across the frequency domain—thereby enabling the pre-configuration of filters to more effectively suppress ambient electromagnetic noise. Furthermore, in contrast to Fig. 16, the detectable frequency range in Fig. 17 extends exclusively above 10 Hz, thereby effectively avoiding the dominant low-frequency interference (< 10 Hz) that severely degrades AMT data quality.

6 Conclusions

Field experiments employing both TFFSM and AMT were conducted in close proximity to a water well within a university campus—a site characterized by intense anthropogenic electromagnetic interference. Based on rigorous analysis of the acquired data, the following conclusions are drawn:

- (1) At locations coinciding with water-bearing geological structures, TFFSM yields distinct low-potential-difference (ΔV) anomalies. Pronounced static shifts are consistently observed in both ΔV profile curves and pseudo-sections. Among the three frequency selectors tested, the TC150 exhibits the most severe static effect.

- (2) Whether each frequency selector employs its dedicated electrode pair or shares a single common electrode pair, the resulting TFFSM measurements remain statistically indistinguishable—indicating negligible influence of electrode configuration on detection fidelity.
- (3) With $MN = 10$ m, site-specific interference significantly degrades TFFSM performance below approximately 20 Hz. Moreover, reducing MN further exacerbates low-frequency instability in TFFSM responses. In contrast, AMT measurements are compromised across the entire frequency band, rendering them unreliable under these conditions.
- (4) Decreasing MN leads to a systematic reduction in both the magnitude of observed ΔV and the effective exploration depth of TFFSM. Notably, when $MN = 2$ m, static effects are nearly eliminated from the recorded signals. This suggests that MN must be carefully optimized according to target depth requirements. Presently, several commercial frequency selectors feature fixed 10-m measurement cables—an inflexible design potentially unsuitable for deep targets (~ 1000 m).
- (5) TFFSM demonstrates exceptional site adaptability: the instrumentation is lightweight and portable, enabling deployment in confined spaces. Owing to its frequency-domain signal acquisition strategy, this method demonstrates superior performance in mitigating environmental electromagnetic interference compared with conventional MT and AMT techniques. Consequently, TFFSM constitutes a viable and practical technique for urban geophysical prospecting.

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Author contributions

T.C. Yang and Z. Yang wrote the main body text. Q. Qin participated the data processing. Y. Hussain improved the English of the text and provided editorial suggestions. Q. Yu and M. Zhu participated in field practice and data interpretation. All authors reviewed and agreed to the published version of the manuscript.

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Data availability

The original contributions presented in this study are included in the article. Further inquiries can be directed to the corresponding author.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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